

Exam-wise Notes — IBPS SO IT Officer (SPL-XVI)

Cloud Computing • Cyber Security • Cryptography | broadened to full CS/IT depth (reusable library topic)

How to read this exam-wise PDF

Covers the topics **unique to IBPS SO IT** beyond the shared umbrella — **Cloud Computing** and **Cyber Security** — written at **full depth** so it doubles as a permanent library topic for *any* future exam testing security/cloud (not just IBPS).

■ **Green box** = how exams test the topic. ■ **Purple box** = scope/pacing note.

For IBPS itself, the depth needed is **conceptual recall** (definitions, models, terms). The deeper cryptography math is marked so you can skim it for IBPS but keep it for harder future exams.

0 What this PDF covers

IBPS SO IT's Professional Knowledge draws mostly from the shared core (CN, OS, DBMS, DS, programming), all in your umbrella. What it adds beyond that — and what this PDF supplies — is modern infrastructure & security:

- **Part A — Cloud Computing:** service models, deployment models, virtualization, containers, key concepts.
- **Part B — Cyber Security:** CIA triad, threats & attacks, defences (firewall/IDS/VPN), authentication.
- **Part C — Cryptography:** symmetric/asymmetric, hashing, digital signatures, PKI, common algorithms.

■ Tested pattern & focus (PSU / Govt CS-IT-Systems)

For IBPS: expect **direct concept MCQs** — service-model definitions (IaaS/PaaS/SaaS), attack-type identification (phishing/DoS/MITM), symmetric vs asymmetric, and 'what does a digital signature provide'.

Highest-yield: **IaaS/PaaS/SaaS**, **CIA triad**, **symmetric vs asymmetric keys**, **hashing vs encryption**, and **malware types**.

PART A Cloud Computing

Cloud computing is on-demand delivery of computing resources (servers, storage, databases, software) over the internet, paid per use. It replaces owning hardware with renting capacity that scales elastically.

A.1 Essential characteristics (NIST)

- **On-demand self-service** — provision resources automatically, no human intervention.
- **Broad network access** — available over the network from any device.
- **Resource pooling** — provider's resources shared (multi-tenant) across users.
- **Rapid elasticity** — scale up/down quickly with demand.
- **Measured service** — pay-per-use metering.

A.2 Service models — the IaaS / PaaS / SaaS ladder

Model	You get / manage	You do NOT manage	Example
IaaS	Virtual machines, storage, network (you manage OS & up)	Physical hardware	AWS EC2, Azure VM
PaaS	A platform to build/deploy apps (you manage code + data)	OS, runtime, servers	Google App Engine, Heroku
SaaS	Ready-to-use software (you just use it)	Everything below the app	Gmail, Salesforce, Office 365

■ The memory hook exams reward

Going IaaS → PaaS → SaaS, **you manage less and the provider manages more**. IaaS = most control, SaaS = least control/most convenience. 'Which model gives a ready-to-use application?' → SaaS.

A.3 Deployment models

Model	Meaning
Public cloud	Shared infrastructure owned by a provider, open to all (AWS, Azure)
Private cloud	Dedicated to one organization; more control & security
Hybrid cloud	Mix of public + private, with data/apps moving between
Community cloud	Shared by several organizations with common concerns

A.4 Enabling technology: virtualization & containers

- **Virtualization** — a **hypervisor** runs multiple **virtual machines (VMs)**, each with its own guest OS, on one physical host. Type 1 (bare-metal) vs Type 2 (hosted) hypervisors.
- **Containers** (e.g. Docker) — lighter than VMs; they share the host OS kernel and package just the app + dependencies. **Kubernetes** orchestrates containers at scale.
- **VM vs container**: VMs virtualize hardware (heavier, full OS each); containers virtualize the OS (lighter, faster, share kernel).

■ Coverage note / warning

For IBPS, the service/deployment-model definitions and the VM-vs-container idea are enough. Deeper cloud architecture (load balancing, auto-scaling groups, serverless/FaaS, CDN) is retained here as **library depth** for future exams — skim for IBPS.

PART B Cyber Security

B.1 The CIA triad — the foundation

Goal	Meaning	Broken by
Confidentiality	Only authorised parties can read data	Eavesdropping, data breach
Integrity	Data isn't altered by unauthorised parties	Tampering, MITM
Availability	Systems/data are accessible when needed	DoS / DDoS attacks

Two more often added: **Authentication** (verifying identity) and **Non-repudiation** (a party can't deny an action — provided by digital signatures).

B.2 Common threats & attacks

Attack / threat	What it is
Malware	Malicious software — umbrella term (below)
Virus	Attaches to a file/program; spreads when run
Worm	Self-replicating; spreads without a host file
Trojan	Disguised as legitimate software
Ransomware	Encrypts data, demands payment
Spyware	Secretly gathers user info
Phishing	Fraudulent messages trick users into revealing data
DoS / DDoS	Floods a system to make it unavailable
MITM	Attacker secretly relays/alters communication
SQL injection	Malicious SQL via input to attack a database
XSS	Injecting malicious scripts into web pages
Brute force	Trying all key/password combinations

B.3 Defences

- **Firewall** — filters traffic by rules at the network boundary (packet-filter, stateful, application/proxy).
- **IDS / IPS** — Intrusion Detection (alerts) / Prevention (blocks) systems.
- **VPN** — encrypted tunnel over a public network for secure remote access.
- **Antivirus, patching, least-privilege access, backups** — baseline hygiene.
- **Authentication factors**: something you know (password), have (OTP/token), are (biometric). **MFA** combines two+.

■ Tested pattern & focus (PSU / Govt CS-IT-Systems)

Near-certain MCQs: **CIA triad** and which attack breaks which goal (DoS → availability); **virus vs worm** (worm needs no host); **phishing** definition; **firewall** purpose; **MFA** factor types.

PART C Cryptography

Cryptography secures information by transforming **plaintext** into unreadable **ciphertext** using an algorithm (**cipher**) and a **key**. **Encryption** scrambles; **decryption** reverses it.

C.1 Symmetric vs asymmetric — the central distinction

	Symmetric (secret-key)	Asymmetric (public-key)
Keys	One shared secret key for both encrypt & decrypt	Key pair: public (encrypt) + private (decrypt)
Speed	Fast — good for bulk data	Slow — good for small data / key exchange
Key problem	Securely sharing the key is hard	Public key shared openly; no sharing problem
Examples	AES, DES, 3DES, Blowfish	RSA, ECC, Diffie-Hellman

■ How they're used together (TLS/HTTPS)

In practice, **asymmetric** crypto securely exchanges a temporary **symmetric** key, then fast symmetric crypto encrypts the actual data. This hybrid is how HTTPS/TLS works — a common exam point.

C.2 Hashing

- A **hash function** maps data to a fixed-size digest; it's **one-way** (can't reverse) and **collision-resistant**.
- Used for **integrity** (verify data unchanged) and **password storage** (store the hash, not the password).
- Examples: **MD5** (broken), **SHA-1** (deprecated), **SHA-256** (current standard).
- **Encryption vs hashing**: encryption is reversible (with a key); hashing is one-way. A frequent MCQ.

C.3 Digital signatures & PKI

- **Digital signature** — the sender **hashes** the message and encrypts the hash with their **private key**; anyone can verify with the sender's **public key**. Provides **authentication, integrity, and non-repudiation**.
- **Digital certificate** — binds a public key to an identity, issued by a **Certificate Authority (CA)**.
- **PKI (Public Key Infrastructure)** — the framework of CAs, certificates, and keys that makes public-key trust work.

■ Tested pattern & focus (PSU / Govt CS-IT-Systems)

High-yield: **symmetric vs asymmetric** (keys, speed, examples), **hashing is one-way**, what a **digital signature provides** (non-repudiation via private-key signing), and **CA/PKI** role.

■ Coverage note / warning

Algorithm internals (DES rounds, RSA modular-exponentiation math, Diffie-Hellman derivation) are kept as **library depth** for GATE/harder exams. For IBPS, the conceptual level above is sufficient.

Rapid recap — IBPS exam-specific (+ reusable)

Area	Nail these
Cloud models	IaaS(infra, most control) → PaaS(platform) → SaaS(software, least control); public/private/hybrid/community deployment
Virtualization	Hypervisor runs VMs (full guest OS each); containers share host kernel (lighter); Docker/Kubernetes
CIA triad	Confidentiality/Integrity/Availability; DoS breaks availability; +authentication +non-repudiation
Attacks	virus(needs host) vs worm(self-spread); trojan, ransomware, phishing, DoS/DDoS, MITM, SQL injection, XSS
Defences	Firewall, IDS/IPS, VPN, MFA (know/have/are factors)
Crypto keys	Symmetric(1 shared key, fast: AES/DES) vs asymmetric(public+private, slow: RSA/ECC); TLS uses both
Hashing/signatures	Hash = one-way (SHA-256), for integrity/passwords; digital signature = private-key sign + public-key verify → non-repudiation; PKI/CA

✓ Where this sits

This completes CS/IT coverage for **IBPS SO IT**: umbrella Modules 1-6 (shared core) + this Cloud/Security/Crypto PDF. Written at full depth, it's also a permanent **library topic** reusable for any future exam testing cloud or security.

Next: the **DSSSB (DOE)** exam-wise PDF (Java/.NET/web/graphics/MIS + IT fundamentals/Python/Pandas), then the shared **Emerging Tech & Security** and future library-filler modules. Q&A sets follow your walkthroughs.